

SSD File:**All_Fires_23_24_gw.shp****IDENTIFICATION INFORMATION****Title:** Fires - All 2023/24**Date of layer completion:** 14/08/2024**Brief Description:** All fires 2023/24**Abstract Description:**

All recorded fires on CapeNature managed property.

Description of Purpose:

The dataset is captured as part of CapeNature's fire management policy to provide an accurate record of the fire history of those areas managed by CapeNature, in order to facilitate ongoing analysis of veld age, fire frequency and fire return intervals as indicators of the state of the veld managed by CapeNature.

Methodology / Lineage Description:

HISTORY: Digitising of historical fire records was initiated by Chris Burgers in 1998, with the assistance of Riki de Villiers. It was substantially added to during 2006-7 by the CAPE GEF Fire project by Granville van Ross and Belinda Munzami. All this work was vetted by and large numbers of further records added by Arne Purves, Helen de Klerk, Cher-Lynn Petersen and Therese Forsyth during 2006-9. Significant contributions were also made by Anne-Lise Schutte-Vlok, Paul Buchholz, Tony Marshall and Patrick Meyer. Our thanks to Armin Seydack who made many fire records available for various reserves in the Southern Cape, and Greg Forsyth making records available for fires on state forests land, that were not available in CapeNature's records.

Since 2000, most Reserve Managers digitize fires themselves using on-screen digitizing over aerial photos and satellite images and GPS (either by walking or flying the fire line) and enter the attributes of the fire report directly into CapeNature's fire database. All records are centralized and collated by the GIS Fire Technician at Scientific Services.

During a follow-up project, extensive data cleaning was done by Therese Forsyth from July 2012 to 2018 to topologically clean the fire scar shapefiles. This entailed overlaying fire scars per fire season and cleaning out duplicates, slivers and overlaps that lead to too short fire intervals. For fires since 2005, the fires were also verified against the annual SPOT5 satellite mosaic images that were made available on an annual basis and since 2016 using satellite imagery made available online. Where possible fire scars were redigitized based on these satellite images.

For the past 5 to 6 years, many large fires were digitized from downloaded georeferenced thumbnail satellite images of Landsat 8 or Sentinel-2. The thumbnails are available from the USGS website. The Sentinel-2 imagery is now also available through CapeFarmMapper where direct digitising of fire scars can be done.

The All_Fires_23_24_gw.shp dataset was created by merging all the individual fire shapefiles into one fire shapefile using "Geoprocessing | Merge". This merged shapefile already contained a column called [FIRE_CODE] as all the individual shapefiles were standardized to have this column. Some basic checks were performed to ensure all individual shapefiles were merged, such as comparing it with the dataset generated in the previous year (using table "Joins") and checking fires reported in past fire season against the fire register.

The columns [Month] (short integer) and [Year] (long integer) were added and the information for these columns were extracted from the fire code using the "Field Calculator, Python" and the script-code [Month] = !FIRE_CODE![5:7] and [Year] = !FIRE_CODE![8:12]. The columns [Res_Code] (text, 5) and [Land_Unit] (text, 50) were added. The [Res_Code] field can be populated using the "Field Calculator, Python" and the script-code [Res_Code] = !FIRE_CODE![0:4]. The [Land_Unit] information was derived from the "CNC_Landscape_units.shp" data layer by using a spatial select (Select by Location) based on the centroid of the fire scar. The column [RESCODE_LU] (text, 10) was added and the landscape populated based on the reserve code and where this reserve centre reports formally.

Further attribute information was extracted to MS Excel spreadsheets from the Central Fire Database. The tables extracted were tblFire.xlsx, tblIgnition.xlsx, and tblIgnitionCause.xlsx. These table were then converted to CSV (comma delimited) files and loaded into ArcGIS. First (i) joined "tblFire.csv" to "All_fires_23_24.shp" using fields [FireReference] to [FIRE_CODE] and exported it to "All_fires_temp1.shp", (ii) then joined "tblIgnition.csv" to "All_fires_temp1.shp" using fields [FireID] to [FireID] and exported it to "All_fires_temp2.shp", and (iii) then joined "tblIgnitionCause.csv" to "All_fires_temp2.shp" using fields [IgnitionCauseID] to [IgnitionCauseID] and exported it to "All_fires_23_24_merge_fields.shp". Deleted the unnecessary fields (FireReference, Firespotte, cnareabur, mcaareabur, otherareab, totalareab, reportcomp, approvedbu, qualityche, featurecou, firespot_1, notes, nearmiss, reservacen, postfiremo, ignitionid, fireid, dds, dde, comparmen, veldtype, veldage, altitude, landowner, tblignitio, causerelia, otherdescr). The layer was then spatially sorted ascending by year using "Data Management tool | General | Sort".

The area in hectare was calculated in ArcGIS.

Product Classification: Standard**Temporal Extent of the Dataset:****Descriptive Keywords:** All fires

Supplemental Information / Report Reference:

Category Name: Natural Environment

Project Name:

Scale:

Bounding Polygon: **North =** -30.42 **East =** 24.22
 South = -34.83 **West =** 17.76

Theme Type: Feature

Content Type: Polygon

Language: English

Minimum Zoom:

Maximum Zoom:

DISTRIBUTION INFORMATION

Path: q:\2_geographic\Natural_Environment\Abiotic\Fires\All_fires\

File Name: All_Fires_23_24_gw.shp

File Type Name: ArcView Shapefile

Distribution - Internal: Yes

External: Yes

List of Formats:

File Size: 18.2mb

Online Resource URL:

Distribution Contact Information and Ordering Instructions

Name of Contact Person: Therese Forsyth

Name of Contact Organisation: CapeNature

Position of Responsible Person: GIS Specialist

Contact Address: Private Bag X29, Gatesville 7766

Contact E-mail: tforsyth@capenature.co.za

Contact Telephone: 087 158 0102

Ordering Instruction:

WCNCB standard: Request the layer via e-mail or in writing from Biodiversity Intelligence, using email datasharing@capenature.co.za. External users will be asked to sign a data agreement.

Fees and Terms:

WCNCB standard: No fees for shapefiles provided by e-mail.

Turnaround Time:

WCNCB standard: Need a minimum of two weeks for delivery.

DATA OWNER AND METADATA INFORMATION

Data Owner

Name of Contact Person: Therese Forsyth

Name of Contact Organisation: CapeNature

Position of Responsible Person: GIS Specialist

Contact Address: Private Bag X29, Gatesville 7766

Contact E-mail: tforsyth@capenature.co.za

Contact Telephone: 087 158 0102

Data Origin from Old records:

Metadata Creator

Metadata Creator Name: Naeem Petersen

Contact Telephone: 087 158 0111

Contact E-mail: npetersen@capenature.co.za

Metadata Date Stamp: 16/08/2024

CONSTRAINTS AND LIMITATIONS

Use Constraints:

WCNCB standard: All data are supplied without distribution rights. You may not redistribute, copy or sell in any form (value added or not) these data to any other parties/bodies outside your organisation/department. All data sets are supplied with metadata files which, inter alia, include information on the custodian of the specific data set. Any reports and/or publications resulting from your project must acknowledge the Western Cape Nature Conservation Board (WCNCB), Scientific Services and, if different, the custodian of the data.

Access Constraints:

Copyright: Yes

Disclaimer:

WCNCB standard: All data are supplied with no expressed or implied warranty as to their suitability for purpose, planimetric accuracy or completeness. The WCNCB cannot be held responsible for any errors, which may occur in provided data sets.

PROJECTION PARAMETERS

Reference System Name:		Reference System Code:	
Projection Name:	Geographic - WGS84	Projection Type:	
Central Meridian:	0	Upper Parallel:	0
Latitude of Projection Origin:		Lower Parallel:	0
Projection Units:		False Easting:	
UTM Zone:		False Northing:	
Datum Name:	WGS84		
Scale Factor at Equator:			

MAINTENANCE INFORMATION

Maintenance and Update Frequency: Annually

Revision and Update History**CONTENT INFORMATION**

Has SAGDAD been used? No

Has another Feature Catalogue been used? No

Catalogue Title: **Catalogue Date:**

Attribute Codes and Descriptions

Field	Description	Alias
Id	Id	
Fire_code	Fire Code (Reserve code/month/year/nr of fire in the month)	
FIREWEB	Fire code assigned within FireWeb app	
Res_code	Four digit reserve code	
Land_Unit	CapeNature landscape unit names (2019), spatial location of centroid of fire scar	
Month	Month (zero value means the month is unknown)	
Year	Year	
Res_Centre	Reserve centre name	
Res_Name	Reserve Name	
Local_Desc	Locality Description	
Date_Start	Start Date of the fire	
Date_Extin	Date the fire was Extinguished	
Date-Withd	Date the team was Withdrawn	
Report_Off	Reporting Officer	
Polic_Case	Police Case Number	
Ignitionca	Ignition Cause	
Yr_mnth	Year / month reference, combination of the year and month values for easy sorting	
Area_ha	Area in hectares calculated in ArcGIS	
RESCODE_LU	Landscape (East, West, Central, South), based on the official split of the reserve centers per landscape	

Thumbnail: